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Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.

2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.

3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.

4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.

5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.

6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.

7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.

9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.

10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

EE1060: Differential Equations and Transform Techniques

Lecture Notes - 27

Topics covered :

1. Impulse Signal
2. Impulse Response

- An impulse $\delta(t)$ can be considered as a signal that when passed through an integrator results in a unit output. Since integrator typically represents an area operator, the impulse signal can be defined as a limiting case of a signal that has unit area i.e. $\delta(t) = \lim_{\epsilon \to 0} s(t)$ where $s(t)$ has unit area.

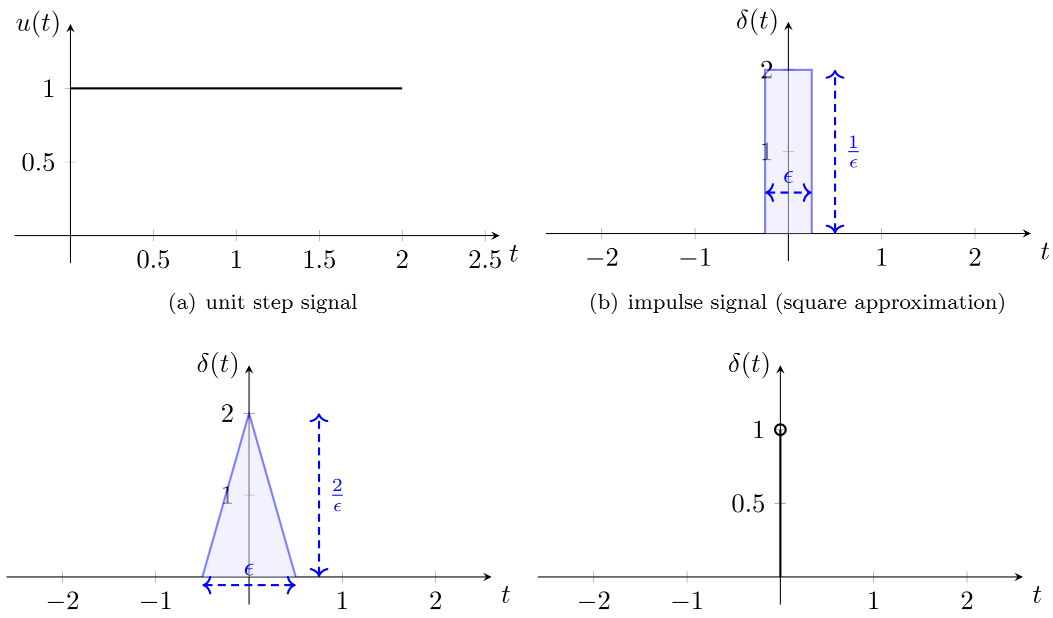
  The two possible signals $s(t)$ that can be used for impulse representation are the rectangular pulse and the triangular pulse (as shown in Fig. 1(b) and Fig. 1(c) respectively). The most commonly used basis signal is the rectangular pulse.

- An impulse signal can be defined as

$$\delta(t) = \lim_{\epsilon \to 0} s(t) \tag{1}$$

  An alternate and most commonly used representation for impulse signals in CT domain is the arrow notation as shown in Fig. 1(d). One point to keep in mind is that impulse signal has an unit area. Furthermore, the impulse function $\delta(t)$ is defined by the property that

$$\int_a^b f(t)\delta(t)\partial t \;=\; f(0) \tag{2}$$



(a) unit step signal (b) impulse signal (square approximation)

(c) impulse signal (triangle approximation) (d) impulse signal (compact representation)

Fig. 1: Unit step signal

with the shifting property resulting in

$$\int_{t=-\infty}^{t=\infty} x(t)\delta(t-\tau)\,dt \;=\; x(\tau) \tag{3}$$

- **Impulse response of a first-order system**: Consider adding a negative feedback to the system. (as shown in Fig. 2).

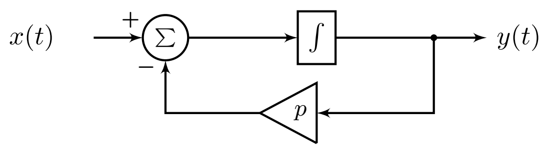
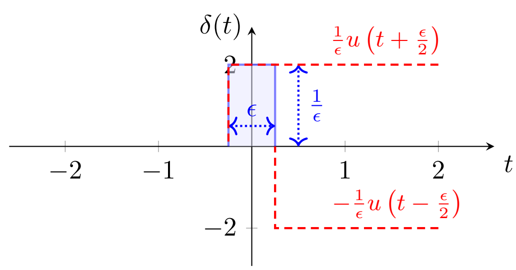


Fig. 2: Block diagram representation of a CT system with feedback

The best way to obtain the impulse response is to go back to the governing equation, which is given by

$$\frac{dy(t)}{dt} + ay(t) \;=\; x(t) \tag{4}$$

The impulse response can be obtained as a limiting case to the response of the system for pulse i.e. the response of the system is first determined for a pulse signal $s(t)$ as shown in Fig. 3 and subsequently, the impulse response is obtained by making $\epsilon \to 0$. The response to the pulse signal can be obtained as a summation of responses to



(a) Impulse approximation

Fig. 3: Steps in computing the response of a CT system with feedback

the two unit step signals i.e.

$$s(t) \;=\; \frac{1}{\epsilon}u\left(t+\frac{\epsilon}{2}\right) - \frac{1}{\epsilon}u\left(t-\frac{\epsilon}{2}\right) \tag{5}$$

Since the system is linear, the response can be simplified by first computing the response to step signal $u(t)$ and then using the time invariance property to obtain the response to $s(t)$. The step response of the system can be computed as [using the solution of first order differential equations by Integrating factor approach]

$$y(t) \;=\; \frac{1}{a}\left(1-e^{-at}\right) \text{ Response to step input} \tag{6}$$

The response to input $s(t)$ can be obtained using time invariant property as

$$\begin{aligned} y(t) &= \frac{1}{a\epsilon}\left[1 - e^{-a(t+0.5\epsilon)} - \left(1 - e^{-a(t-0.5\epsilon)}\right)\right] \\ &= \frac{e^{-at}}{a\epsilon}\left[e^{\frac{p\epsilon}{2}} - e^{-\frac{p\epsilon}{2}}\right] \end{aligned} \tag{7}$$

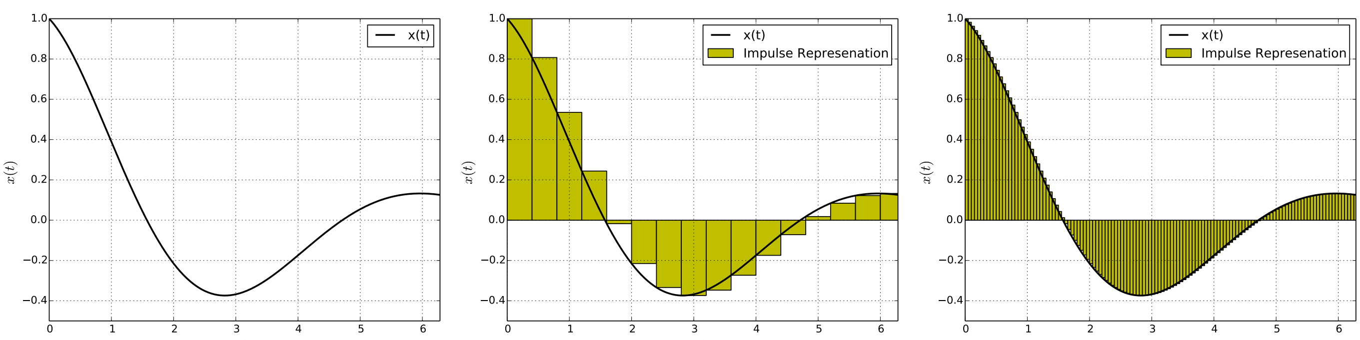
The impulse response (obtained by taking the limiting $y(t)$ as $\epsilon \to 0$) can be computed to be

$$h(t) \;=\; \lim_{\epsilon\to 0} y(t) \;=\; e^{-at} \tag{8}$$

The impulse response of CT system with negative feedback decays to zero as long as $a > 0$ and will increase if $a = 0$. On the other hand the impulse response is constant when $a = 0$. Similar to DT systems, the output of the feedback system is characterized by the gain a which is referred to as **Pole** of the system. Further, the analysis shows that CT systems of the form shown in Fig. 2 are stable if and only if the pole $a > 0$.

- **Synthesizing a signal using impulses**: For linear systems the response to any input can be obtained using the principle of superposition. The principle of superposition states that if the input can be expressed as the sum of signals, the output can be expressed as the sum of the responses to the respective signals.

  The first step in employing the superposition theorem for obtaining response of linear system to an arbitrary input is to decompose the input into the sum of impulse signals. To start with, the input signal is represented using a rectangular impulse of width Δ and subsequently the representation in terms of impulses by setting $\Delta \to 0$. Any signal $x(t)$ can be represented using impulses as indicated in Fig. 4.



(a) Original signal $x(t)$

(b) Representation of $x(t)$ using impulses of width 0.40

(c) Representation of $x(t)$ using impulses of width 0.05

Fig. 4: Representation of $x(t)$ using rectangular impulse

From Fig. 4, it can be inferred that an input signal can be written in terms of rectangular impulses as

$$x(t) = \sum_{k=-\infty}^{k=\infty} x(k\Delta)\delta_{\Delta}(t - k\Delta)\Delta \tag{9}$$

An observation into figure can indicates that an accurate representation can be obtained using making $\Delta \to 0$. As

$\Delta \to 0$, the $\sum$ becomes an integral and the $\delta_\Delta(t - k\Delta)$ becomes an impulse δ. So,

$$x(t) = \int_{\tau=-\infty}^{\tau=\infty} x(\tau)\delta(t-\tau)\partial\tau \tag{10}$$

One must not get confused the above representation of input signal with the definition of impulse function. In impulse representation of input signal, the input signal is represented by a sequence of time delayed impulses whose magnitude is $x(\tau)$ and the value of τ sweeps from $-\infty$ to ∞.